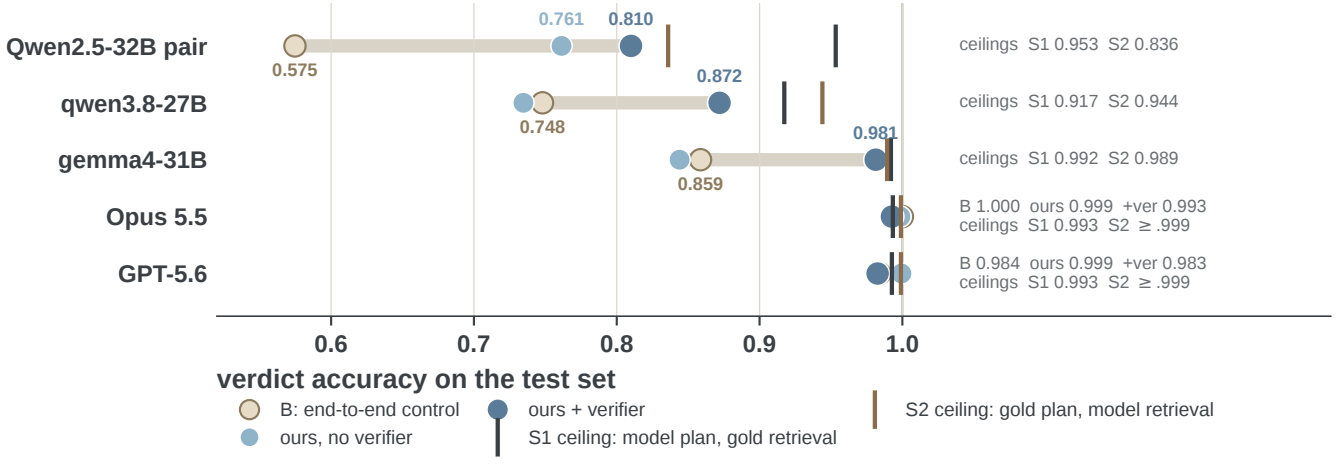
Qwen2.5-32B pair
0.761
0.810
0.575
ceilings S1 0.953 S2 0.836
qwen3.8-27B
0.872
0.748
ceilings S1 0.917 S2 0.944
gemma4-31B
0.981
0.859
ceilings S1 0.992 S2 0.989
Opus 5.5
B 1.000 ours 0.999 +ver 0.993
ceilings S1 0.993 S2 ≥ .999
GPT-5.6
B 0.984 ours 0.999 +ver 0.983
ceilings S1 0.993 S2 ≥ .999
0.6
0.7
0.8
0.9
1.0
verdict accuracy on the test set
B: end-to-end control
ours, no verifier
ours + verifier
S1 ceiling: model plan, gold retrieval
S2 ceiling: gold plan, model retrieval